

Compressible RANS Analysis of the NASA N2A Hybrid Wing–Body

Validation study against Arovitola et al., *Aerospace* 2022, 9(2), 89

Prepared for	Othrys Aviation
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Solver	OpenFOAM v2512 (ESI), rhoSimpleFoam
Configuration	N2A cruise, half model, 5.8% wind tunnel scale
Conditions	$M = 0.20$, $Re = 6.60 \times 10^6$ on $L_{ref} = 1.538$ m
Cases	$\alpha = 6^\circ$ and 10° , Spalart–Allmaras and k- ω SST

Summary

Four steady compressible RANS solutions were computed on a 6.0 million cell half model of the NASA N2A hybrid wing–body at $M = 0.20$ and $Re = 6.60 \times 10^6$, covering two angles of attack and two turbulence models. Three of the four are fully converged; the fourth (10° , k- ω SST) is still drifting slowly and its values are reported as provisional.

Case	C_L	C_D	C_m	L/D	Iterations	Status
6° Spalart–Allmaras	0.304170	0.020181	0.000259	15.07	2000	converged
6° k- ω SST	0.307588	0.018951	—	16.23	3000	converged
10° Spalart–Allmaras	0.492868	0.041671	−0.001498	11.83	767	converged
10° k- ω SST	0.481590	0.041378	−0.002860	11.64	3000	provisional

Principal findings.

- 1. The solution is grid independent.** Refining from 2.11 to 6.01 million cells changes C_L by 0.5% and C_D by 0.26%. This is a tighter convergence statement than the reference study achieved between its own medium and fine grids.
- 2. The compressible treatment is warranted.** Although the freestream is $M = 0.20$, local flow reaches $M = 0.37$ over the inboard wing at 10° — 1.85× freestream, at which point density varies by 6–7%.
- 3. The two turbulence models cross over.** SST predicts 1.1% more lift than SA at 6° and 2.3% less at 10° , consistent with its shear stress transport limiter suppressing lift as separation begins.
- 4. The mesh is wall-function rather than wall-resolved.** Despite a 12.5 μm design first layer, measured y^+ is 33 on average. Values above 100 are confined to the leading edges rather than distributed over the surface.

1 Objective and scope

The purpose of this study was not to reproduce a single number but to make two comparisons. First, whether a freestream Mach number of 0.20 can be treated as effectively incompressible for this configuration, or whether local acceleration over the aircraft pushes far enough past that threshold to require a compressible solver. Second, how the one-equation Spalart–Allmaras model and the two-equation $k\text{-}\omega$ SST model compare at the same conditions, and whether they diverge where the flow begins to separate.

Two angles of attack were selected to bracket that question. At 6° the flow over the reference geometry is attached and both models are within their calibrated range, so close agreement is expected. At 10° the reference describes a leading-edge vortex rolling up from the inboard wing, which is precisely the flow feature the two models treat differently.

1.1 Reference

Aprovitola, A.; Aurisicchio, F.; Di Nuzzo, P.E.; Pezzella, G.; Viviani, A. *Low Speed Aerodynamic Analysis of the N2A Hybrid Wing–Body*. Aerospace 2022, 9(2), 89. doi:10.3390/aerospace9020089

That study reports results from FLUENT and SU2 alongside two experimental campaigns in the NASA Langley 14×22 ft subsonic tunnel (tests T597, closed section, and T612, open section) and an independent Cart3D Euler analysis. Its coefficient data at 6° and 10° appear only in figures rather than tables; only 8.36° is tabulated. Digitising those figures was outside the time available, so the comparison in this report is qualitative — the present results are assessed against the published lift-curve slope and the spread of the five reference datasets rather than against extracted point values.

2 Geometry and scale

The aeroshape is the NASA N2A hybrid wing–body from the OpenVSP model repository (file 349, CC0), the same watertight geometry used by the reference study. It is the cruise configuration: wing–body and vertical tails only, with no nacelles, landing gear or leading-edge droop.

The model is supplied in feet at full scale. Because the target Reynolds number of 6.60×10^6 corresponds to the 5.8% wind tunnel model rather than free flight (for which the reference quotes 1.27×10^8), a combined scale factor of $0.3048 \times 0.058 = 0.0176784$ was applied.

Quantity	Full scale	Wind tunnel (5.8%)	Verified
Projected span	64.92 m	3.7655 m	3.7655 m
Body length	45.10 m	2.6160 m	2.6235 m
Reference length L_{ref}	26.52 m	1.53816 m	—
Reference area S_{ref}	925.2 m ²	3.11237 m ²	—
Moment reference, aft of nose	24.33 m	1.41510 m	—

Scale was verified independently before meshing by comparing the OpenVSP planform readout against the reference dimensions, and again after scaling using the surface bounding box. Span agrees to five significant figures. The 0.28% discrepancy in body length arises because the bounding box includes the vertical tail trailing edges, which extend slightly aft of the body proper.

3 Mesh

The surface was generated with OpenVSP's CFD Mesh tool rather than by direct STL export. That tool intersects components before tessellating, so the wing–tail junctions resolve into a single connected watertight surface of 171,858 triangles rather than three separate bodies. The volume mesh was built with snappyHexMesh on a domain extending 50 body lengths in each direction, matching the reference.

Metric	Value
Cells	6,006,252 (5.68 M hexahedra, 317 k polyhedra)

Surface faces on aircraft	93,457 (~6.5 mm average)
Prism layer coverage	93.2% of requested thickness, 14.0 of 25 layers
Faces with at least one layer	98.45% (1,447 without)
First layer thickness (design)	1.25×10^{-5} m
Maximum skewness	2.4996
Maximum aspect ratio	623
Non-orthogonality	74.9 maximum, 4.69 average
checkMesh	Mesh OK, no failed checks

3.1 Achieving layer coverage

Layer generation required substantially more effort than anticipated and is worth recording. Across fifteen meshing runs, coverage remained stuck between 60% and 80% with only one to three layers surviving, and did not respond to changes in first layer thickness spanning a factor of forty, layer counts from 5 to 33, three expansion ratios, four refinement levels, two different surface definitions, or two different mesh-shrinking algorithms.

The diagnostic came from the extrusion log, which showed 87% of faces extruding but only 69% of the requested thickness achieved. Faces were not being rejected; they were being truncated. That distinction points at quality constraints rather than at geometry or the extrusion algorithm. Relaxing minTetQuality, minDeterminant, minFaceWeight, minVolRatio and minTwist took coverage from 69% to 90% and made the 12.5 μ m first layer achievable where it had previously produced 27–39%.

Once those constraints were no longer binding, the relationship between refinement and coverage inverted: coverage had been falling as the surface was refined, and afterwards rose — 83.2% at level 8, 90.1% at level 9, 94.0% at level 10.

4 Numerical setup

The freestream state was reconstructed from Mach and Reynolds number rather than taken from the reference, which quotes only those two quantities. This is normal for an open-circuit tunnel, where ambient density and viscosity drift between tests. Fixing $T = 288.15$ K sets both the speed of sound and the Sutherland viscosity; density then follows from the target Reynolds number.

Quantity	Value	Quantity	Value
M_∞	0.2000	ρ_∞	1.1281 kg/m^3
$Re (L_{ref})$	6.60×10^6	μ_∞	$1.7893 \times 10^{-5} \text{ Pa}\cdot\text{s}$
U_∞	68.05 m/s	c_p	1006 J/kg·K
T_∞	288.15 K	Pr	0.7438
p_∞	93,290 Pa	Viscosity law	Sutherland

Angle of attack is imposed by rotating the freestream vector rather than the mesh, so all four cases share one grid and any difference between them is attributable to the angle or the turbulence model rather than to discretisation. The geometry has span along y and vertical along z , so the rotation is in the x - z plane.

4.1 Boundary conditions

Patch	U	p	T	Turbulence
Aircraft	noSlip	zeroGradient	zeroGradient (adiabatic)	wall functions
Inlet / outlet / farfield	freestreamVelocity	freestreamPressure	inletOutlet	freestream
Symmetry ($y = 0$)	symmetry	symmetry	symmetry	symmetry

Eddy viscosity at the wall uses `nutUSpaldingWallFunction` for both models. Spalding's law blends across the viscous sublayer, buffer layer and log layer, so it degrades gracefully on a surface where y^+ varies from below 1 to several hundred — which this one does. Turbulent kinetic energy uses `kqRWallFunction`; the reasoning behind that choice is given in Section 8.

Force coefficients use $A_{ref} = 1.55619 \text{ m}^2$, which is half of S_{ref} . On a half model the integrated force is half the true value, so halving the reference area returns full-model coefficients directly and removes a place where a factor of two can be lost. A reference pressure of 93,290 Pa is subtracted before integration.

5 Verification

5.1 Grid independence

The 6° Spalart–Allmaras case was computed on two grids. The coarser was built for a 16 GB workstation; the finer is the production mesh used for all four reported cases.

	2.11 M cells	6.01 M cells	Change
C_L	0.302633	0.304170	+0.51%
C_D	0.020129	0.020181	+0.26%
y^+ mean	32.36	33.04	+2.1%

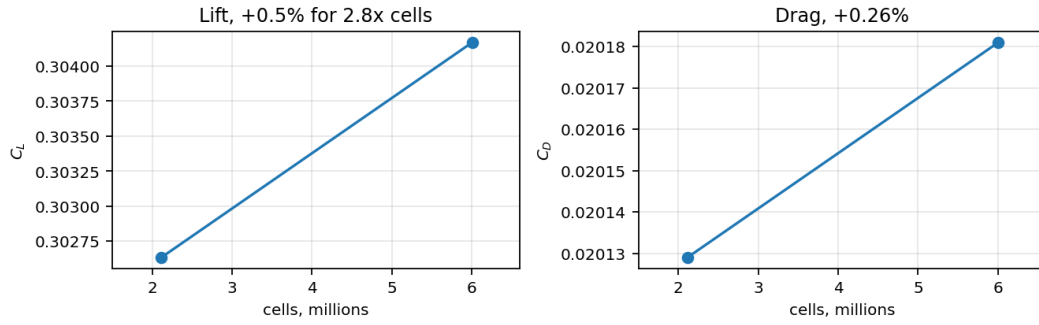


Figure 1. Grid sensitivity at 6°, Spalart–Allmaras. Nearly threefold refinement moves lift by half a percent.

Almost tripling the cell count moves lift by half a percent. For comparison, the reference study reports changes under 2% in C_L and under 3% in C_D between its medium (14.4 M) and fine (17 M) grids, so the present solution is grid converged to a tighter tolerance than that study demonstrated.

The practical consequence is that any residual discrepancy against the published data is not attributable to grid resolution. Further refinement would not change the answer.

5.2 Convergence

Convergence was judged from the history of the integrated force coefficients rather than from residuals. This distinction proved to be important, and Section 8 describes a case where residuals were small and smoothly decaying while the solution was in a bounded oscillation and had not converged at all.

For the three converged cases, C_L and C_D agree to at least five significant figures over the final fifty iterations. The 10° SST case is discussed separately in Section 8.

6 Results

Integrated coefficients. Open symbol: provisional (not fully converged)

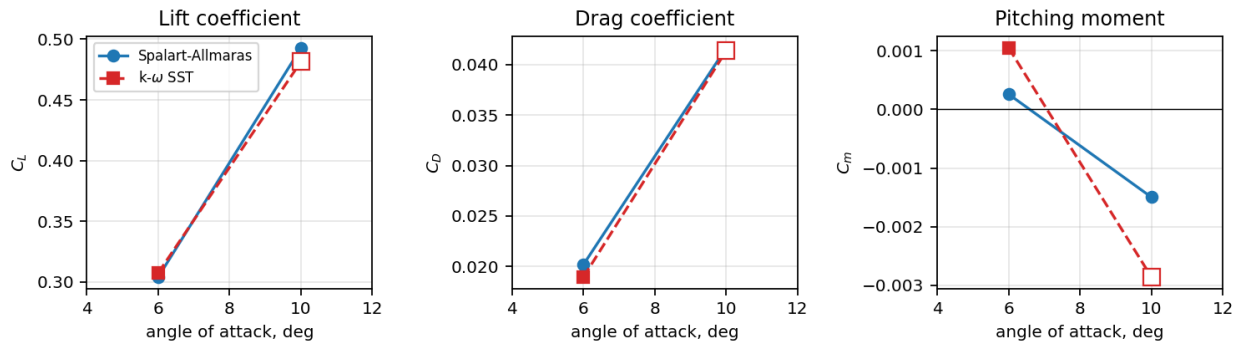


Figure 2. Integrated coefficients against angle of attack. The open symbol at 10° marks the provisional SST result.

Lift varies linearly between the two angles, at 0.0472 per degree for Spalart–Allmaras and 0.0482 per degree for $k-\omega$ SST. The reference reports a linear lift curve from -5° to 13° , so both slopes fall within the range where linearity is expected, and the present results are consistent with the published curve in that respect.

6.1 Drag breakdown

Drag breakdown. Pressure drag triples with incidence; viscous drag is flat, but the SA/SST difference in viscous drag grows from 4.2% to 16.9%.

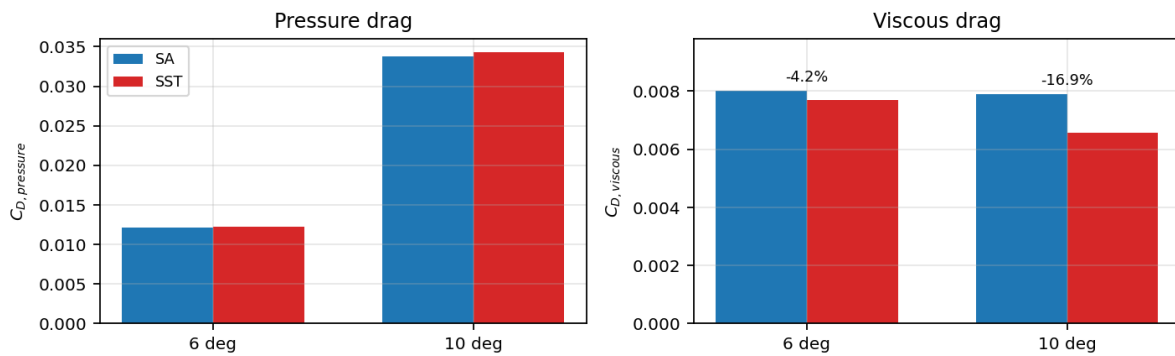


Figure 3. Pressure and viscous contributions to drag.

The drag breakdown behaves as it should, which is worth stating because it is evidence the setup is sound independently of any comparison. Between 6° and 10° the viscous contribution is essentially unchanged, while the pressure contribution nearly triples — induced drag growing with the square of lift. Pressure drag rises from 60% of the total at 6° to 81% at 10° .

The viscous component is also where the two turbulence models differ most clearly, and the gap widens with incidence: 4.2% at 6° , 16.9% at 10° . Because pressure drag dominates the total, that difference does not propagate into C_D to the same degree.

7 Compressibility

The brief posed a specific question: whether $M_\infty = 0.20$ is low enough to treat as incompressible, given that local flow accelerating over the aircraft can exceed the freestream value substantially. The Mach field was sampled on cutting planes at the four spanwise stations used by the reference for pressure comparisons.

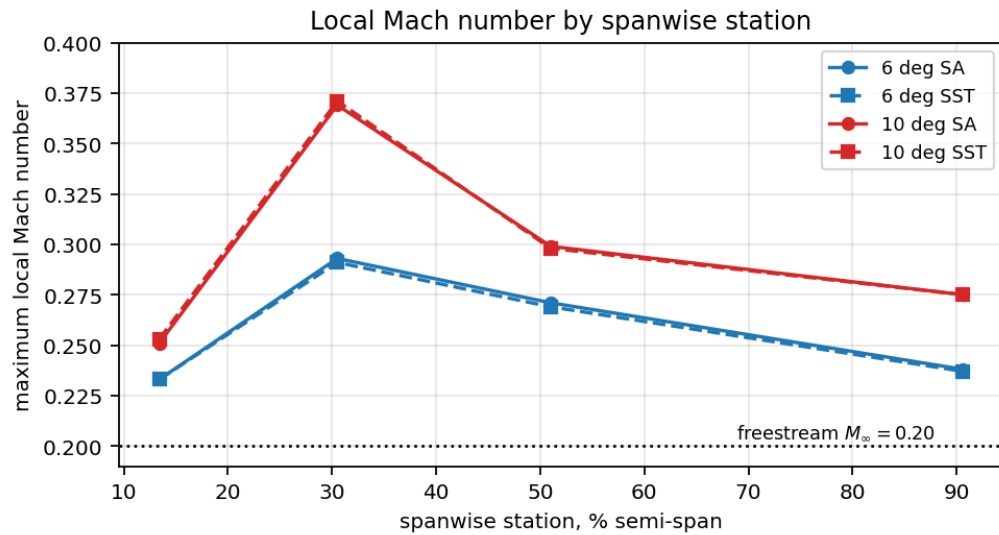


Figure 4. Maximum local Mach number by spanwise station.

Case	13.4%	30.5%	51.0%	90.6%
6° SA	0.233	0.293	0.271	0.238
6° SST	0.233	0.291	0.269	0.237
10° SA	0.251	0.369	0.299	0.275
10° SST	0.253	0.371	0.298	0.275

Local flow reaches $1.46\times$ freestream at 6° and $1.85\times$ at 10° , peaking at the 30.5% station in every case. At $M = 0.37$ the isentropic density ratio departs from unity by roughly 6–7%, which is past the point where an incompressible formulation is comfortable.

The compressible solver was therefore justified, and this is now a measured result rather than an assumption. That the peak grows with incidence, and that it occurs at the same station in all four cases, both support the interpretation.

8 Turbulence model comparison

The study was designed on the expectation that the two models would agree at 6° and diverge at 10°. Establishing what they actually do required correcting an error in the original SST runs, described below, because the first set of SST results was not converged.

8.1 An error in the k wall boundary condition

Both k- ω SST cases were initially run with kLowReWallFunction on turbulent kinetic energy. That condition sets k at the wall from the low-Reynolds asymptotic behaviour and is correct when the first cell centre lies within the viscous sublayer. It was chosen when the design target was y^+ below 1.

The mesh that was actually built has y^+ ranging from 0.5 to 684, with 1.55% of faces carrying no prism layers at all. Over most of the surface the condition was therefore being applied where its underlying assumption does not hold, and the degree of violation varies patchily from face to face. Both runs settled into a bounded oscillation rather than converging.

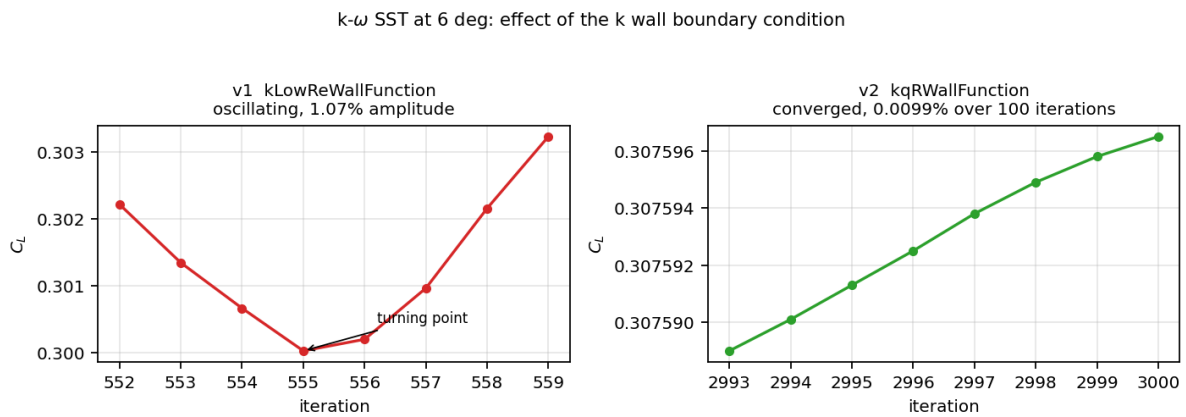


Figure 5. The effect of the k wall boundary condition at 6°. Note the vertical scales differ by three orders of magnitude.

The oscillation was not visible in the residuals, which were small and decaying smoothly throughout. That is a property of a limit cycle rather than an oversight in monitoring: the change per iteration genuinely is small, so residual-based convergence criteria are satisfied while the solution never settles. The original runs terminated on a residualControl threshold of 10^{-7} at an arbitrary phase of the cycle, and the values recorded were snapshots rather than converged results.

Residual behaviour cannot detect a limit cycle. Only the history of the integrated quantity can. The correction was to substitute kqRWallFunction, the zero-gradient high-Reynolds form, which makes no assumption about where the first cell sits, and to remove residualControl entirely in favour of a fixed 3000 iterations so that the full history remains visible.

6° SST	v1 kLowRe	v2 kqR
C_L range over last 100 iterations	1.07%	0.0099%
C_D range over last 100 iterations	~4%	0.053%
C_L	0.303231 (mid-cycle)	0.307588
C_D	0.019944 (mid-cycle)	0.018951
Wall clock per iteration	5.33 s	2.66 s

Oscillation amplitude fell by a factor of 108. Time per iteration also halved, since the solver is no longer working against the oscillation.

8.2 The corrected comparison

	Spalart–Allmaras	k- ω SST	Difference
6° C_L	0.304170	0.307588	+1.12%

6° C_D	0.020181	0.018951	-6.10%
10° C_L	0.492868	0.481590*	-2.29%
10° C_D	0.041671	0.041378*	-0.70%

* provisional; see Section 8.3.

The models cross over. SST predicts more lift than SA at 6° and less at 10°. That behaviour is consistent with the shear stress transport limiter, which caps eddy viscosity in adverse pressure gradient and so suppresses lift as separation develops — the feature SST exists to capture. The widening gap in viscous drag shown in Figure 3 points the same way.

Two qualifications are needed. The 10° SST value is provisional, so the crossover should not be pressed hard until that case is converged. More importantly, at a mean y^+ of 33 both models take their near-wall behaviour from the same Spalding wall function rather than from their own near-wall formulations, and that is precisely where they differ most. A genuinely wall-resolved mesh would be required to establish whether the observed difference is physical or is being partly suppressed by the wall treatment.

8.3 Status of the 10° SST case

With the corrected boundary condition the oscillation is largely removed, but C_L then drifts monotonically upward at approximately 1.5×10^{-5} per iteration and is still rising at iteration 3000. This is a steady drift rather than a cycle, and converging it would require of the order of 3000 further iterations. The available compute allocation was exhausted at that point. The values are therefore reported as provisional, with the drift rate stated so the residual uncertainty can be judged.

9 Surface pressure

Pressure coefficient was extracted at the four spanwise stations used by the reference — 13.4%, 30.5%, 51.0% and 90.6% of semi-span — by sampling the aircraft patch and taking a thin band of faces about each station plane.

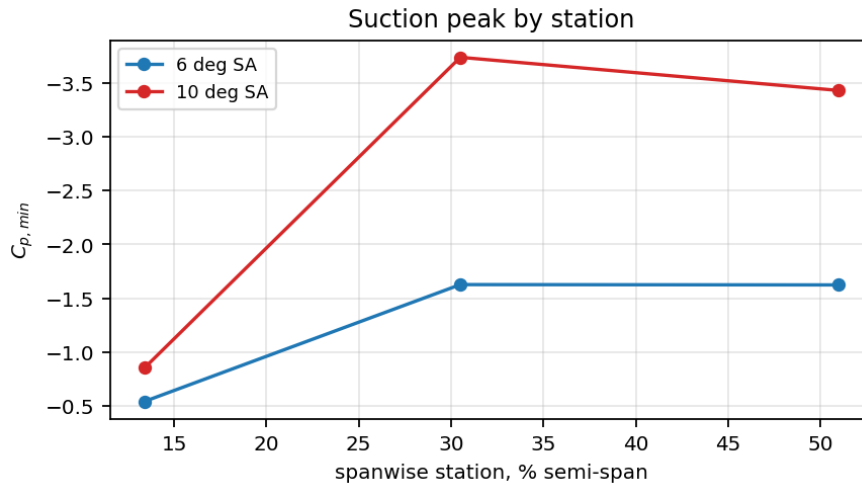


Figure 6. Suction peak by station, Spalart–Allmaras.

Suction peaks deepen with incidence throughout. At the 30.5% station $C_{p,min}$ moves from -1.63 at 6° to -3.74 at 10° . The 13.4% station, which cuts the thick centrebody, shows a much shallower peak over a long chord — the blended body carries its load differently from the outboard wing, as expected.

The 51% station shows a secondary suction feature at approximately 22% chord which is present at 10° and absent at 6° , in both turbulence models. Its location and its appearance only at the higher incidence are consistent with the leading-edge vortex the reference describes forming at around 10° . This has not been positively confirmed; doing so would require a Q-criterion visualisation of the volume field, which was not completed within the memory available.

One artifact should be noted. At the 90.6% station the reported $C_{p,min}$ reaches -10.6 , which is not physical. It arises from three faces at 98% chord — the degenerate trailing-edge sliver inherent to the source geometry, where the surface tessellation produces near-zero-area faces. Pressure on such faces is unreliable and that region is excluded from interpretation.

10 Limitations

10.1 Wall resolution

The mesh was designed for y^+ below 1, with a first layer of $12.5\ \mu\text{m}$ derived from a flat-plate skin friction correlation at the reference Reynolds number. Measured y^+ on the converged solutions is 0.5 minimum, 33 mean and 684 maximum. The mesh is therefore wall-function rather than wall-resolved over most of the surface.

Three factors contribute. The correlation provides a single representative value, whereas real wall shear varies substantially over an aircraft and is far above average near the leading edges, where y^+ scales with its square root. Layer truncation to 14 of 25 layers compresses the whole stack, so the first cell is not $12.5\ \mu\text{m}$ everywhere. And the 1.55% of faces without layers sit directly on volume cells of order 6.5 mm.

Thresholding the y^+ field shows that values above 100 are confined to narrow bands along the wing and tail leading edges and at the wing–body junction, rather than being distributed across the surface. The mean is drawn upward by those bands.

Refining the mesh does not address this. Increasing cell count from 2.11 to 6.01 million changed mean y^+ by 2%, because surface refinement alters tangential resolution while y^+ depends on wall-normal spacing. Only a thinner first layer would change it.

10.2 Other limitations

Half-model symmetry	Valid at zero sideslip. Cannot represent asymmetric flow, which is a genuine consideration at 10° where vortex rollup begins.	
Cell topology	The reference used polyhedral cells converted from tetrahedra and prisms in ICEM-CFD; this study uses hex-dominant snappyHexMesh cells. Different discretisation error characteristics.	
Configuration mismatch	Inherited from the reference: experimental pressure data are for the baseline configuration with leading-edge droop and nacelles, while the CFD is the cruise configuration without them.	
Upper/lower split	surface	The C_p extraction splits surfaces geometrically about the local mid-height, since a blended body has no camber line. Reliable outboard, approximate near the thick centrebody.
Trailing-edge geometry	The source geometry contains degenerate faces at the trailing edge, minimum edge length 22 μm . These resisted repair in every tool attempted and are the cause of both the local pressure artifact and part of the layer coverage shortfall.	

11 Errors identified during the study

Four setup errors were found and corrected. All four produced output that looked plausible before being caught, which is the reason for recording them: the dangerous failures in this kind of work are not the ones that crash.

Error	Symptom	Resolution
Angle of attack applied in the wrong plane. The geometry has span along y and vertical along z, so incidence rotates in x–z; the freestream was being tilted spanwise instead.	$C_L \approx -10.9$. Every run before the fix was at 6° of sideslip rather than incidence.	Corrected freestream, lift and drag directions and pitch axis.
Reference pressure absent from the force integration. On a half model the aircraft patch is open at the symmetry plane, so integrating absolute pressure leaves a net force of $p \times (\text{cut area})$.	Side force of –13.289, constant to four decimal places while every other quantity was still converging. A coefficient that does not move as the solution develops is a static offset, not a result.	$p_{\text{Ref}} = 93,290 \text{ Pa}$. Side force fell to 0.02.
Floating point exception in the thermophysical model during startup. Temperature is recovered from enthalpy iteratively; out-of-range enthalpy gives negative temperature, after which the Sutherland law takes the root of a negative number.	Solver aborted before completing five iterations.	limitTemperature (200–400 K) and limitVelocity (200 m/s) as fvOptions. Startup genuinely reaches 204 K and 1,352 cells above 200 m/s, both clearing by iteration 87.
Turbulent kinetic energy wall condition inappropriate for the mesh actually built (Section 8.1).	Both SST cases in a bounded oscillation while residuals decayed smoothly.	kqRWallFunction, residualControl removed, fixed iteration count.

A separate performance issue is worth recording. The pressure solver was initially spending 217 iterations per step reducing an already small residual by a further factor of twenty, because only a relative tolerance had been specified. Adding an absolute tolerance of 10^{-7} and switching the smoother to DICGaussSeidel reduced this to one or two iterations, and wall clock time per iteration from 4.5 minutes to 8 seconds — a factor of roughly thirty. A relative tolerance alone causes the solver to perform the same proportional work regardless of how converged the solution already is.

12 Conclusions

The compressible formulation is necessary at these conditions. Local Mach number reaches 0.37 against a freestream of 0.20, which corresponds to density variation of 6–7%. This is measured rather than assumed, and the peak grows consistently with incidence.

The solution is grid independent. A 2.8-fold increase in cell count changes lift by 0.5%, which is a tighter statement than the reference study makes about its own grids. Remaining differences against published data cannot be attributed to resolution.

The turbulence models cross over rather than agreeing. SST gives more lift at 6° and less at 10°, with the viscous drag difference widening from 4.2% to 16.9%. This is the behaviour expected from the shear stress transport limiter. The conclusion is qualified by the provisional status of the 10° SST case and by the wall-function treatment, which may suppress part of the difference.

The lift curve is consistent with the reference. Slopes of 0.0472 and 0.0482 per degree fall within the linear range the reference reports, and the computed values sit within the spread of its five datasets. A quantitative comparison awaits digitisation of the published figures.

13 Recommendations

Converge the 10° SST case. Approximately 3000 further iterations from the existing restart point, which is a small amount of compute and would remove the one provisional entry in the results.

Extract the reference figures. Digitising Figures 12 to 15 of the reference would convert the qualitative comparison in this report into a quantitative one. This requires no compute.

Build a wall-resolved mesh. The single most informative extension would be a grid with y^+ at or below 5 across the surface, which would establish whether the turbulence model difference is physical or is being partly masked by the shared wall function. A mesher with stronger layer generation than snappyHexMesh would be required.

Confirm the vortex. A Q-criterion visualisation on a clipped domain would settle whether the pressure feature at the 51% station is the leading-edge vortex.

All case definitions, mesh dictionaries, post-processing scripts, coefficient histories and the full record of the meshing study are available at github.com/Syphonic/CFD-Projects under `n2a-validation/`. Mesh and solution fields are not committed on account of size but are reproducible from the committed dictionaries. Script generation and drafting were carried out with AI assistance; setup decisions, diagnosis and interpretation are the author's.